

Route-F P54: Common Complex Phases, Finite Pati–Salam Matching, and the Self-Consistency Interface

P54PQ-v2 four-dimensional mainline

5 September 2026

Abstract

We transport the actual four-copy scalar eigenvector into a single standard-hypercharge spinor convention. This fixes the up/down conjugation, the relative Majorana/triplet phases, and the interference of the two seesaw terms. The previous magnitude-only light-copy dictionary is superseded. We also construct a nearby Pati–Salam-symmetric stationary saddle of the same frozen action, whose integrated 55-real-dimensional scalar block is positive. It yields a finite upper gauge threshold, including the vector constant. Retained tachyons do not invalidate this heavy-field matching. The lower effective-action matching and finite Yukawa diagrams must still be supplied before a physical global fit is defined. Numerical algebraic regressions and synthetic family choices are not experimental fits.

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1 Scope and the fixed action

The action, field content and scalar benchmark remain P54PQ-v2: $3\,16_F + 54_{H,\mathbb{R}} + 126_{H,\mathbb{C}} + 10_{H,\mathbb{C}} + 1_{H,\mathbb{C}}$, with PQ charges $(-1, 0, +2, -2, -4)$ in the order $(\Psi, \Phi, \Sigma, \phi, S)$. No extra family matrix, scalar multiplet, torsion sector or lattice scan is introduced. The previous checkpoint was committed and pushed as **b8e7398**; the calculations below are the subsequent checkpoint.

We retain 328 canonical real scalar coordinates and the fixed-VEV background-field Landau- $\overline{\text{MS}}$ curvature prescription from the companion **p54_fixed_vev_full_doublet_matching.tex**. Numerical masses below are in units of the historical $\omega = 1$. The old physical values of M_I, M_U remain unrecertified: the earlier threshold census was inconsistent with its PS beta functions. New dimensionless matching tensors do not by themselves supply replacement physical scales.

2 One complex scalar–spinor dictionary

2.1 Construction from Clifford intertwiners

On five qubits define

$$\Gamma_{2j-1} = Z^{\otimes(j-1)} \otimes X \otimes I^{\otimes(5-j)}, \quad \Gamma_{2j} = Z^{\otimes(j-1)} \otimes Y \otimes I^{\otimes(5-j)}, \quad (1)$$

$$\Gamma_* = (-i)^5 \Gamma_1 \cdots \Gamma_{10}, \quad C = \Gamma_2 \Gamma_4 \Gamma_6 \Gamma_8 \Gamma_{10}. \quad (2)$$

They obey $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}$ and $C^T = -C$. The restrictions of $C\Gamma_i$ and $C\Gamma_{[5]}$ to the appropriate chiral subspace are symmetric, as required by two left-handed Weyl fields. For an antisymmetric vector generator R ,

$$S(R) = \frac{1}{2} \sum_{i < j} R_{ij} \Gamma_i \Gamma_j, \quad S(R)^T B_a + B_a S(R) + \sum_b B_b R_{ba} = 0. \quad (3)$$

The five-form version uses the induced wedge representation. The numerical audit tests all 45 generators, not only the hypercharge eigenvalues.

Let U_{126} be the existing orthonormal ordered-five-form embedding. Its old orientation selects the conjugate standard-matter hypercharges. We make an explicit global coordinate change, rather than conjugating an isolated Clebsch coefficient:

$$x_{\text{phys}} = K x_{\text{old}}, \quad U_{126, \text{phys}} = \overline{U}_{126}, \quad \Gamma_* \Psi_{\text{phys}} = +\Psi_{\text{phys}}. \quad (4)$$

Here K flips the imaginary real-coordinate blocks of Σ, ϕ, S and leaves all other coordinates unchanged. Consequently

$$V_{\text{phys}}(x) = V_{\text{old}}(Kx), \quad H_{\text{phys}} = K H_{\text{old}} K, \quad B_{\text{phys}} = K B_{\text{old}}. \quad (5)$$

If complex old-action couplings had been fixed, their conjugation is transported too. In the new raw convention

$$-\mathcal{L}_Y = \frac{1}{2} h_{ij} \Psi_i^T C \Gamma_a \Psi_j \phi_a^* + \frac{1}{2 \cdot 5!} f_{ij} \Psi_i^T C \Gamma_{a_1 \dots a_5} \Psi_j \Sigma_{a_1 \dots a_5} + \text{h.c.} \quad (6)$$

The physical spinor has hypercharge multiset $(1/6)^6, (-2/3)^3, (1/3)^3, (-1/2)^2, 1, 0$.

The actual old copy order is retained:

$$B = (B_{\phi, \text{hol}}, B_{\phi, \text{anti}}, B_{\Sigma, \text{hol}}, B_{\Sigma, \text{anti}}), \quad x_{\text{phys}} = 2 \text{Re}(B_{\text{phys}} c H^0), \quad \langle H^0 \rangle = v/\sqrt{2}. \quad (7)$$

All right-handed-conjugate component phases are fixed once by requiring the corresponding reference 10_H coefficient to be positive. The same states are then used for every 126_H , Majorana and triplet entry. Direct matrix-element contraction gives

	h_{raw}	f_{raw}
Y_u	$\sqrt{2} c_1$	$2c_4/\sqrt{3}$
Y_d	$\sqrt{2} c_2^*$	$2c_3^*/\sqrt{3}$
Y_e	$\sqrt{2} c_2^*$	$-6c_3^*/\sqrt{3}$
Y_ν	$\sqrt{2} c_1$	$-6c_4/\sqrt{3}$

(8)

These are matrix coefficients: each entry multiplies its raw family matrix. The c versus c^* distinction is required by H^0 versus H^{0*} . It cannot be inferred from $|c_a|$.

2.2 The corrected bosonic light map

Define Dirac-unit couplings $h_D = \sqrt{2}h_{\text{raw}}$ and $f_D = 2f_{\text{raw}}/\sqrt{3}$. Equation (8) implies

$$Y_d = ah_D + df_D, \quad Y_e = ah_D - 3df_D, \quad (9)$$

$$Y_u = bh_D + ef_D, \quad Y_\nu = bh_D - 3ef_D, \quad (a, b, d, e) = (c_2^*, c_1, c_3^*, c_4). \quad (10)$$

At the fixed-VEV bosonic-retuned point,

$$a = -0.7631652014906 - 1.03 \times 10^{-12}i, \quad b = 0.6462011667549, \quad (11)$$

$$d = 1.2183583687513 \times 10^{-5} + 5.47 \times 10^{-15}i, \quad e = -0.001710897393866 - 1.02 \times 10^{-14}i. \quad (12)$$

For nonzero a, b, d the conventional ratios are

$$r = b/a = -0.846738249455, \quad s = ae/(bd) = 165.843989136. \quad (13)$$

The same dictionary at the tree point gives $r = -0.905014358012$, $s = -29.7554251165$. In particular the small coefficient d crosses zero under the bosonic correction. A large change in s , which divides by d , is not a large rotation of the entire scalar eigenvector. Numerically stable fits must use (a, b, d, e) directly, not (r, s) near $d = 0$. Earlier reciprocal/magnitude-only copy maps, and bounds using those maps, are not current physical constraints. This is not a claim that flavor tension has disappeared, nor a whole-model exclusion.

2.3 Majorana and triplet phases in the same basis

The canonical singlet and LL-triplet contractions are

$$\kappa_R = +i4\sqrt{2}, \quad \kappa_{LL} = -i4\sqrt{2}, \quad f_M = \kappa_R f_{\text{raw}} = i2\sqrt{6}f_D, \quad M_R = \sigma\omega f_M. \quad (14)$$

Here the numerical $\sigma = 0.1265$ is dimensionless and ω restores mass units. The i is a component-basis convention, not an additional physical CP parameter. Dropping it in only one term would nevertheless change the declared matching map.

The existing two-triplet tree Hessian and cubic source, evaluated along the bosonic-improved light direction, induce

$$\frac{\omega\Delta_{126}}{v^2} = -0.003323921052868. \quad (15)$$

The 54 triplet has zero renormalizable LL coupling but contributes to this response through its mixing with 126. Contracting the full response in the common dictionary yields

$$M_\nu = M_L - \frac{v^2}{2} Y_\nu M_R^{-1} Y_\nu^T \quad (16)$$

$$= \frac{v^2}{\omega} [C_{II,D} f_D + C_{I,D} Y_\nu f_D^{-1} Y_\nu^T], \quad (17)$$

$$C_{II,D} = 0.01628382104964i, \quad C_{I,D} = \frac{-1}{2\sigma i 2\sqrt{6}} = 0.80681480328827i. \quad (18)$$

This formula requires invertible f_D . If a neutrino is not hard at the matching scale it must remain in the EFT; the inverse is not to be regularized by hand. The scalar response is still mixed-order (tree K, J with bosonic one-loop c), not complete one-loop Weinberg matching.

For independent neutrino basis changes $\nu_L = U_L \nu'_L$, $\nu^c = U_R \nu'^c$, one has

$$Y'_\nu = U_L^T Y_\nu U_R, \quad M'_R = U_R^T M_R U_R, \quad M'_L = U_L^T M_L U_L, \quad (19)$$

so the full seesaw transforms as $M'_\nu = U_L^T M_\nu U_L$. The proof follows by inserting $(U_R^T M_R U_R)^{-1} = U_R^\dagger M_R^{-1} U_R^*$. The verifier also rotates actual scalar copies by random $U(4)$ and tests the conjugate down rule. Deliberately replacing down c^* with c gives an order-one residual, so the regression is not insensitive to this error.

3 A controlled upper PS finite gauge threshold

3.1 Same-action stationary saddle and heavy-field elimination

Retain the four complex PS parents

$$(1, 2, 2)_{10}, \quad (15, 2, 2)_{126}, \quad (\overline{10}, 3, 1)_{126}, \quad (10, 1, 3)_{126}. \quad (20)$$

These are the common standard-matter labels, with $4 = (3, 1/3) + (1, -1)$ under $SU(3) \times U(1)_{B-L}$. The old literal-P1 labels are conjugated by the dictionary change. The LL scalar has $B-L = +2$ and its singlet triplet has $Y = +1$; the right-handed-conjugate pair instead requires scalar $B-L = -2$. Their Casimirs and real index traces are unchanged. Their real dimension is $8 + 120 + 60 + 60 = 248$. At $\sigma = 0$ solve the two heavy radial equations

$$\partial_\omega V(\omega, 0, v_s) = 0, \quad \partial_{v_s} V(\omega, 0, v_s) = 0 \quad (21)$$

with every existing parameter, including the three quadratic masses, unchanged. This gives

$$\omega_{PS} = 1.0008103059, \quad v_{s,PS} = 0.2542776971. \quad (22)$$

The full gradient vanishes. There are 24 upper gauge Goldstones and one PQ direction. The complement of these and the active 248-plane has dimension 55 and satisfies

$$\lambda_{\min}(H_{HH}) = 0.10288204 > 0. \quad (23)$$

All 124 negative directions are retained active fields, rather than modes being integrated out. Thus this point is a stationary saddle, not a proposed stable vacuum of the full theory.

Proposition 1 (Local heavy-field EFT). *On a fixed gauge slice, let h be heavy and ℓ retained coordinates. If $V_h(h_0, \ell_0) = 0$ and $V_{hh} > 0$, the implicit-function theorem yields a unique local heavy valley $h = h_*(\ell)$ and a real local tree EFT $V_{\text{EFT}}(\ell) = V(h_*(\ell), \ell)$. No positivity assumption on the retained Hessian is needed.*

Proof. Invertibility of V_{hh} applies the implicit-function theorem to $V_h = 0$. Differentiating gives $h'_* = -V_{hh}^{-1}V_{h\ell}$. At a stationary point,

$$(V_{\text{EFT}})_{\ell\ell} = V_{\ell\ell} - V_{\ell h}V_{hh}^{-1}V_{h\ell}. \quad (24)$$

The positive heavy block fixes the local branch and its real Gaussian functional. Tachyons in ℓ are represented by the EFT itself. $\square$

This is local existence, not a global heavy valley or a proof that a renormalizable truncation captures all powers of σ/ω .

3.2 Finite constants and logarithmic consistency

Use real-representation index operators normalized to $\text{Tr } I_i = T_i(R_{\mathbb{R}})$. For the present upper threshold, with no massive charged fermions removed, the gauge matching is [5]

$$\alpha_i^{-1}(\mu) = \alpha_{10}^{-1}(\mu) - \frac{\lambda_{U,i}(\mu)}{12\pi}, \quad (25)$$

$$\lambda_{U,i} = \text{Tr}_{V_H} I_i - 21 \text{Tr}_{V_H} \left[I_i \log \frac{M_V}{\mu} \right] + \text{Tr}_{S_H} \left[I_i \log \frac{M_S}{\mu} \right]. \quad (26)$$

Goldstone directions are removed once from the physical scalar trace; the massive vector coefficient includes the corresponding gauge/ghost sector. The finite vector constant cannot be recovered by matching logarithmic slopes alone. At the PS saddle the active, mass and Goldstone projectors commute. Thus the heavy traces are genuine spectral restrictions.

At $\mu_U = g_U \omega_{PS}$, in group order $(4, L, R)$,

$$\lambda_U = (2.944052689, 7.602978830, 7.602978830), \quad \lambda_U^{V, \text{const}} = (4, 6, 6). \quad (27)$$

The upper physical real scalar indices are $(12, 6, 6)$, so

$$\lambda'_U = 21(4, 6, 6) - (12, 6, 6) = (72, 120, 120). \quad (28)$$

These equal $-6(a_{10} - a_{PS})$ with $a_{10} = -34/3$ and $a_{PS} = (2/3, 26/3, 26/3)$, as follows directly by differentiating the inverse-coupling matching equation. The largest retained absolute squared mass divided by the smallest integrated squared mass is 1.01260824: there is no uniform hierarchy between every retained and integrated scalar. Positivity defines the local heavy functional, but does not prove a controlled two-site renormalizable truncation for all intermediate loop momenta.

At the lower site the repaired census fixes, in SM order $(3, 2, 1)$,

$$I_V = (1, 0, 14/5), \quad I_S = (67, 71, 67), \quad \lambda'_I = (-46, -71, -41/5). \quad (29)$$

The hypercharge projection is $a_1^{PS} = 2a_4/5 + 3a_R/5$. These three lower slopes, together with the three upper slopes, reproduce all six beta discontinuities. This does not calculate the finite lower constants.

At the simultaneous broken vacuum the old active projector P_A obeys

$$\|[P_A, H]\|_F \simeq 0.265322, \quad \|[P_A, P_{G_U}]\|_F \simeq 0.609960, \quad \text{Tr}(P_A P_{G_U}) \simeq 0.189003. \quad (30)$$

These are structural violations, not eigensolver rounding. A trace of $P_A \log H$ there cannot replace the lower EFT determinant. As a separate check, the complete one-step $SO(10) \rightarrow SM$ hard spectrum gives

$$\lambda_{\text{one step}} = (36.63010816, -6.54752280, 119.73803708), \quad \lambda' = (26, 49, 92.6). \quad (31)$$

It is a full-spectrum regression, not an adopted change from the two-stage physical branch. Finite shifts of tree masses from the one-loop tadpole counterterms inside a one-loop threshold enter at higher order; all such order choices and the background prescription must remain common when the full finite EFT is assembled.

3.3 The actual lower heavy valley is gapped, but is not a mass-log shortcut

Transport the upper heavy 55-plane to the actual broken vacuum. Let its orthonormal basis be B_H , let B_A span the retained parent plane, and remove the upper orbit from the latter:

$$B = (I - P_{G_U})B_A, \quad C = B_H^T H B_H, \quad A = B^T H B, \quad W = B_H^T H B, \quad E = -C^{-1}W. \quad (32)$$

The calculation verifies $B_H^T B = 0$, SM covariance, and $\lambda_{\min}(C) = 0.099450257411$. The tree heavy-valley action has

$$S = A - W^T C^{-1}W, \quad G = B^T B + E^T E. \quad (33)$$

The generalized problem $Sv = m^2 Gv$ has no negative eigenvalues and exactly 13 zero directions (nine lower Goldstones and four real Higgs components). Its Ward residual is 3.21×10^{-16} , and

$$\lambda(G) \subset [0.984249788817, 1.01800727050], \quad \|E\|_2 = 0.134191171461. \quad (34)$$

This verifies a local derivative-expansion interface on the actual vacuum. It does not make these generalized eigenvalues full pole masses. Indeed, the exact Euclidean quadratic Schur operator is

$$\mathcal{K}_{\text{EFT}}(p^2) = p^2 G_0 + A - W^T (p^2 I + C)^{-1} W, \quad G_0 = B^T B, \quad (35)$$

$$= S + p^2 G - p^4 W^T C^{-3} W + O(p^6). \quad (36)$$

At the operator level $\det \mathcal{K}_{\text{full}} = \det \mathcal{K}_{HH} \det \mathcal{K}_{\text{EFT}}$. Gauging this identity also generates covariant vertices and commutators; replacing it by $\text{Tr} \log S$ loses those terms. The implemented two-state, equal-gauge-charge counterexample distinguishes the exact Schur determinant from a projector-weighted mass logarithm. Completing the lower finite matching means using the induced action, not merely another spectral assignment rule.

4 Complete four-parent one-loop PS Yukawa system

4.1 Derivation and normalization from the actual tensors

Write the real-scalar Weyl Lagrangian as $-\mathcal{L}_Y = \frac{1}{2}\mathcal{Y}_{IJ}^a\psi_I\psi_Jq_a + \text{h.c.}$, where each full $\mathcal{Y}^a$ is symmetric. For the two-component convention, the general one-loop coefficient is [3]

$$16\pi^2\beta_{\mathcal{Y}^a} = \frac{1}{2}(S^T\mathcal{Y}^a + \mathcal{Y}^aS) + 2\sum_b\mathcal{Y}^b\mathcal{Y}^{a\dagger}\mathcal{Y}^b + \sum_b\mathcal{Y}^b\text{Re Tr}(\mathcal{Y}^{b\dagger}\mathcal{Y}^a) - 3\sum_i g_i^2(C_i^T\mathcal{Y}^a + \mathcal{Y}^a C_i), \quad S = \sum_b\mathcal{Y}^{b\dagger}\mathcal{Y}^b. \quad (37)$$

Every complex scalar supplies a pair $\mathcal{Y}^{\text{Re}} = \mathcal{Y}/\sqrt{2}$ and $\mathcal{Y}^{\text{Im}} = \pm i\mathcal{Y}/\sqrt{2}$. For each internal pair, $\mathcal{Y}^{\text{Re}}\mathcal{Y}^{a\dagger}\mathcal{Y}^{\text{Re}} + \mathcal{Y}^{\text{Im}}\mathcal{Y}^{a\dagger}\mathcal{Y}^{\text{Im}} = 0$. This cancels the vertex sum without deleting any scalar parent.

The actual 124 complex PS intertwiners are projections of the common Clifford tensors, not arbitrary Clebsch tables. In a PS fermion order (ψ_L, ψ_R) , a bidoublet family matrix H enters opposite off-diagonal blocks as H and H^T ; thus the full Weyl tensor is symmetric even when H itself is not. The canonical PS normalizations at the tree upper boundary are

$$H = \sqrt{2}h_{\text{raw}}, \quad F = 4f_{\text{raw}}, \quad L = R = 2\sqrt{2}f_{\text{raw}}. \quad (38)$$

In particular $F = 2\sqrt{3}f_D$ and $f_M = i2R$ in the common phase convention. Different normalizations in a reference must be converted before importing coefficients. The Δ_R -only system in Ref. [4] is a comparison limit, not this active census.

Define

$$A_L = HH^\dagger + \frac{15}{4}FF^\dagger + \frac{15}{2}LL^\dagger, \quad (39)$$

$$A_R = H^\dagger H + \frac{15}{4}F^\dagger F + \frac{15}{2}R^\dagger R, \quad (40)$$

$$G_D = \frac{9}{4}(g_L^2 + g_R^2 + 5g_4^2), \quad G_L = \frac{9}{4}(2g_L^2 + 5g_4^2), \quad G_R = \frac{9}{4}(2g_R^2 + 5g_4^2). \quad (41)$$

The complete matrix equations are

$$16\pi^2\dot{H} = A_L H + H A_R + [4\text{Tr}(H^\dagger H) - G_D]H, \quad (42)$$

$$16\pi^2\dot{F} = A_L F + F A_R + [\text{Tr}(F^\dagger F) - G_D]F, \quad (43)$$

$$16\pi^2\dot{L} = A_L L + L A_L^T + [2\text{Tr}(L^\dagger L) - G_L]L, \quad (44)$$

$$16\pi^2\dot{R} = A_R^T R + R A_R + [2\text{Tr}(R^\dagger R) - G_R]R, \quad \dot{} = d/d \log \mu. \quad (45)$$

Here $L = L^T, R = R^T$ are preserved. Away from the appropriate parity locus, H, F are general complex matrices; projecting their antisymmetric parts away at every step would change the differential equation. The actual 248-real-scalar, 48-Weyl-component tensors reproduce Eq. (45) component by component for unequal left/right inputs.

The associated Yukawa contribution to the two-loop gauge flow is

$$Y_4 = (2\text{Tr}(A_L + A_R), 4\text{Tr } A_L, 4\text{Tr } A_R), \quad \dot{g}_i = \frac{g_i^3 a_i}{16\pi^2} + \frac{g_i^3}{(16\pi^2)^2} \left[\sum_j b_{ij} g_j^2 - Y_{4,i} \right], \quad (46)$$

with

$$a = (2/3, 26/3, 26/3), \quad b = \begin{pmatrix} 3551/6 & 249/2 & 249/2 \\ 1245/2 & 779/3 & 48 \\ 1245/2 & 48 & 779/3 \end{pmatrix}. \quad (47)$$

For example the $SU(4)$ factor follows from $C_4(4) = 15/8$, eight left and eight right gauge components, and $\dim SU(4) = 15$ in $\text{Tr}(C_i S)/\dim G_i$. The implementation checks this trace using the actual tensors and also includes a Standard-Model top-only normalization regression.

4.2 A protected one-loop submanifold, not new independent matrices

There is a useful exact simplification of the frozen tree boundary. Suppose

$$g_L = g_R, \quad H = H^T, \quad F = F^T, \quad L = R = F/\sqrt{2}. \quad (48)$$

Then $A_R = A_L^T$, $G_D = G_L = G_R$, and $\text{Tr } F^\dagger F = 2 \text{Tr } R^\dagger R$. Inserting these identities into Eq. (45) gives

$$\beta_H = \beta_H^T, \quad \beta_F = \beta_F^T, \quad \beta_F = \sqrt{2}\beta_L = \sqrt{2}\beta_R. \quad (49)$$

The gauge beta functions also give $\beta_{g_L} = \beta_{g_R}$, since their rows and $\text{Tr } A_L, \text{Tr } A_R$ agree on this locus. Uniqueness of the ODE therefore proves invariance of (48). This is a conditional one-loop protection, not merely a numerical coincidence.

It follows that the tree Spin(10) boundary does *not* acquire three independent arbitrary 126 family matrices just by PS running. On this locus the ratio $f_M/f_D = i2\sqrt{6}$ is preserved at one loop. For finite matching that moves the boundary off the locus, unequal left/right thresholds, decoupling, or higher-loop evolution, the general system must instead transport F, L, R separately. The lower dictionary is always

$$Y_d = aH + \frac{d}{2\sqrt{3}}F, \quad Y_u = bH + \frac{e}{2\sqrt{3}}F, \quad M_R = i2\sigma\omega R, \quad M_L = -i2\Delta_L L, \quad (50)$$

with the same lepton -3 replacements. All four PS matrices remain correlated outputs of two UV matrices plus calculated thresholds. They are not four newly fitted free matrices.

5 Finite complex kinetic matching: an executable interface

5.1 Canonical normalization and the scalar curvature

In the Weyl convention $\psi_L^T Y_a \psi_R z_a$, write the hard local kinetics as $Z_L = I + K_L$, $Z_R = I + K_R$, $Z_H = I + K_H$ and the vertex as $\tilde{Y}_a = Y_a + \delta Y_a$. If these Hermitian metrics are positive, set $S_j = Z_j^{-1/2}$. Direct field substitution gives

$$Y_a^{\text{can}} = \sum_b S_L^T \tilde{Y}_b S_R (S_H)_{ba}, \quad (51)$$

$$D_{\text{can}} = S_H^\dagger D S_H, \quad (52)$$

$$Y_a^{\text{can}} = Y_a + \delta Y_a - \frac{1}{2} \left[K_L^T Y_a + Y_a K_R + \sum_b Y_b (K_H)_{ba} \right] + O(\hbar^2). \quad (53)$$

The linear formula is the general canonical matching algebra; see also Appendix A of Ref. [1]. Its model-specific finite matrices must still be computed from UV-minus-EFT amplitudes. SU(5) group factors from that paper are not P54 coefficients.

If $Dc = 0$ and $c^\dagger c = 1$, then

$$c_{\text{can}} = \frac{Z_H^{1/2} c}{\sqrt{c^\dagger Z_H c}}, \quad D_{\text{can}} c_{\text{can}} = 0, \quad S_H c_{\text{can}} = \frac{c}{\sqrt{c^\dagger Z_H c}}. \quad (54)$$

Thus at an exact zero the original-coordinate light direction changes only by kinetic normalization. An apparent rotation in canonical coordinates is not an additional independent scalar-mixing correction. Vertices and curvature must be transformed together to see this cancellation. For down/e vertices multiplying z^* , use Z_H^* and c^* , not the up formula. The executable tests use the actual four-copy embedding and curvature with synthetic complex finite matrices. They verify independent left/right family transformations, arbitrary $U(4)$ copy rotations, and the conjugate-copy rule.

5.2 Stable finite kernels and what they do not calculate

For squared masses $x, y \geq 0$, at least one nonzero, define

$$f(x, y; \mu) = -\frac{1}{16\pi^2} \int_0^1 \log \frac{(1-t)x + ty}{\mu^2} dt, \quad (55)$$

$$h(x, y; \mu) = \frac{1}{16\pi^2} \int_0^1 (1-t) \log \frac{(1-t)x + ty}{\mu^2} dt. \quad (56)$$

Integrating gives, for unequal positive masses,

$$16\pi^2 f = 1 - \frac{x \log(x/\mu^2) - y \log(y/\mu^2)}{x - y}, \quad (57)$$

$$16\pi^2 h = \frac{1}{2} \log(x/\mu^2) + \frac{\frac{1}{2}r^2 \log r - \frac{3}{4}r^2 + r - \frac{1}{4}}{(1-r)^2}, \quad r = y/x. \quad (58)$$

The integral form avoids subtractive cancellation near equality. At $x = y$, $(16\pi^2 f, 16\pi^2 h) = (-\log(x/\mu^2), \frac{1}{2} \log(x/\mu^2))$. At $y = 0$ they become $(1 - \log(x/\mu^2), \frac{1}{2} \log(x/\mu^2) - \frac{1}{4})$; at $x = 0$ the last constant is $-3/4$. Also

$$\frac{d(f, h)}{d \log \mu} = \frac{(2, -1)}{16\pi^2}. \quad (59)$$

These kernels agree with Appendix B of Ref. [1] and are implemented with equal/unequal/one-zero mass regressions. They are building blocks, not the complete tensor contractions of P54 diagrams. The API rejects missing finite vertices, nonpositive kinetic metrics and purely light kernels rather than filling absent physical inputs with zeros. Using exact square roots of one-loop input matrices is an algebraic regression, not a calculation of two-loop terms.

6 Local scalar–flavor feedback and its analytic derivative

6.1 An actual Clebsch-constrained inner solve

Two deterministic small complex symmetric raw family choices test the inner scalar calculation. They are not measured or globally fitted matrices. The actual copy dictionary supplies all $Y_{\nu, a}$ and $M_R = \kappa_R \sigma f_{\text{raw}}$ in $\omega = 1$ units. With $X = M_R^\dagger M_R$ and $W = X[\log(X/\mu^2) - 1]$, use

$$\Pi_{F, ab} = -\frac{1}{8\pi^2} \text{Tr}(Y_{\nu, a}^\dagger Y_{\nu, b} W), \quad (60)$$

$$\delta\nu_F^2 = -\frac{1}{8\pi^2 \sigma^2} \text{Tr}\{X^2[\log(X/\mu^2) - 1]\}, \quad (61)$$

$$\Delta_F = \Pi_F - \frac{1}{2} \delta\nu_F^2 P_{126}, \quad (62)$$

$$D(\theta, \xi) = D_0 + \Delta_B + \Delta_F(\theta) - \xi P_{10}. \quad (63)$$

The subtraction scale is held fixed in all field derivatives. The scalar ξ is tuned to the first stable zero of the complete complex matrix, not just to a scalar projection. At a gapped zero $Dc = 0$, $c^\dagger c = 1$, choose $c^\dagger c' = 0$. Differentiation gives

$$\xi' = \frac{c^\dagger \Delta_F' c}{c^\dagger P_{10} c}, \quad c' = -D^+(\Delta_F' - \xi' P_{10})c. \quad (64)$$

Here D^+ is the inverse on the heavy complement, zero on the light direction. The formula is valid while the heavy gap and $c^\dagger P_{10} c$ are nonzero.

For the explicit test $f_{\text{raw}}(\theta) = (1 + \theta)f_{\text{raw}}$,

$$X' = 2X, \quad W' = 2X \log(X/\mu^2), \quad (\delta\nu_F^2)' = -\frac{\text{Tr}\{X^2[4 \log(X/\mu^2) - 2]\}}{8\pi^2 \sigma^2}. \quad (65)$$

Apply the product rule also to both $Y_{\nu,a}$ in Π_F to obtain Δ'_F without a finite difference. Then Eq. (64) agrees with two independent finite-difference steps of the retuned eigenpair. Representative results are

case	ξ	heavy gap/ ω^2	$\ dc/d\theta\ _2$
0	-0.02025044324	0.03026151664	0.0008092458
1	-0.02024289530	0.03026452622	0.0012951028

The small-step relative error of c' is about 6×10^{-9} . This validates a differentiable local inner solver. It neither selects physical family parameters nor replaces the finite PS map.

6.2 Recompute the type-II source when the light direction changes

The previous type-II number cannot be kept fixed while c changes. Using the quartic action, reconstruct the exact cubic jet tensor

$$T_{Apq} = V'''[e_A, q_p, q_q], \quad J_A(c) = \sum_{pq} T_{Apq} c_{\mathbb{R},p} c_{\mathbb{R},q}, \quad c_{\mathbb{R}} = (\text{Re } c, \text{Im } c). \quad (66)$$

Centered differences of the Hessian are algebraically exact cubic jets for a quartic potential. Eight already cached Hessians give the four real Higgs-direction jets; hypercharge covariance supplies the four imaginary ones. This introduces no new parameter or lattice scan and reproduces the old source to 4.4×10^{-16} relative error. For each new eigenvector recompute

$$z_{\mathbb{R}}(c) = -\frac{1}{2} K_{\text{tree}}^{-1} J(c), \quad \Delta_A(c) = \frac{z_{R,A} + i z_{I,A}}{\sqrt{2}}, \quad C_{II,\text{raw}}(c) = \sum_A \kappa_{LL,A} \Delta_A(c). \quad (67)$$

For the two examples $|\Delta C_{II,\text{raw}}|$ relative to the old bosonic direction is 2.295×10^{-5} and 3.623×10^{-5} . All inputs, including the cubic tensor, triplet Hessian, source and LL coefficient, are transported in the CP-conjugation test. This tests covariance, not CP invariance of a chosen complex benchmark. The triplet Hessian and cubic vertices are still tree quantities; their complete loop correction, evolved L/R inputs off the protected locus, and finite Weinberg matching remain necessary for a physical fit.

7 Research interpretation and completion boundary

The executable checkpoint passes the following tests on a fresh replay:

Verifier suffix (all under <code>route_f/code/</code>)	passed
<code>common.yukawa.phase</code>	28/28
<code>ps.finite.thresholds</code>	25/25
<code>ps.yukawa.flow</code>	29/29
<code>finite.yukawa.interface</code>	17/17
<code>self.consistent.light</code>	36/36

The full filenames begin with `verify_p54_` and end in `.py`; each exports a same-named JSON/Markdown ledger, with input hashes and individual residuals, under `route_f/output/`. The phase-annotated action card separately passes its existing 20 checks. These 135 new regression assertions are not 135 independent empirical tests of P54. Family choices and finite-array examples remain synthetic.

The recent analyses of Refs. [1, 2] motivate checking finite heavy thresholds before turning a tree Yukawa relation into an exclusion. Their field content and fitted parameters are not those of P54; neither their coefficients nor their viability conclusions are imported. Here the calculable advance is a common phase dictionary and a valid upper heavy-field matching branch, with explicit algebra for transporting finite corrections into the light state.

The remaining physical map must contain both-site finite gauge and Yukawa matching, the appropriate active-field evolution, sequential neutrino and Weinberg matching, and same-action scalar feedback. An optimization over an undefined map is neither a global fit nor evidence that the fit domain is empty. The code therefore distinguishes missing prerequisites from a computed infeasible point.

References

- [1] K. M. Patel and S. K. Shukla, *Quantum corrections and the minimal Yukawa sector of $SU(5)$* , [arXiv:2310.16563v2](#) (2024), especially Appendices A and B.
- [2] S. K. Shukla, *Revisiting $SU(5)$ Yukawa Sectors Through Quantum Corrections*, [arXiv:2411.06906v2](#), revised 25 August 2026.
- [3] M. Luo, H. Wang and Y. Xiao, *Two-loop Renormalization Group Equations in General Gauge Field Theories*, [arXiv:hep-ph/0211440](#).
- [4] D. Meloni, T. Ohlsson and S. Riad, *Renormalization Group Running of Fermion Observables in an Extended Non-Supersymmetric $SO(10)$ Model*, [arXiv:1612.07973](#).
- [5] L. J. Hall, *Grand Unification of Effective Gauge Theories*, [Nucl. Phys. B 178 \(1981\) 75](#).